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Film capacitor, film, and metallized film

Abstract

A film capacitor that includes a wound body having a dielectric film and a metal layer. When a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface of the dielectric film, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of International application No. PCT/JP2021/036981, filed Oct. 6, 2021, which claims priority to Japanese Patent Application No. 2020-171218, filed Oct. 9, 2020, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to a film capacitor, a film, and a metallized film.

BACKGROUND OF THE INVENTION

(2) As a capacitor, a film capacitor having a structure including a flexible film as dielectric film with a first metal layer and a second metal layer opposed to each other across the film is known. The film capacitor is produced, for example, by winding or laminating a film with a first metal layer and a film with a second metal layer.

(3) In production of a film capacitor through fabrication of a wound body by winding a film, the wound body may be pressed for height reduction of the film capacitor. On this occasion, a film having good slidability allows the wound body to be uniformly pressed, so that the height reduction of the film capacitor can be easily achieved.

(4) On the other hand, it is said that formation of a gap between the overlapping films in a wound body allows decomposition gas from the film to easily scatter from inside the film capacitor in dielectric breakdown, so that a so-called self-healing function works to recover the insulating state of the film. In fabrication of a wound body, a film having good slidability allows a gap to be uniformly formed between the overlapping films, so that the self-healing function works easily.

(5) Therefore, in order to enhance the pressing properties against a film capacitor and self-healing properties of a film capacitor, slidability is imparted to a film in some cases. As a method for imparting slidability to a film, a method of blending an organic filler into a base resin is disclosed in Patent Literature 1. As described above, slidability is conventionally imparted to a film by blending a filler into a resin to roughen the film surface. Patent Literature 1: JP 2011-251493A

SUMMARY OF THE INVENTION

(6) In a conventional film with a filler blended into a resin, however, an electric field tends to be concentrated in the vicinity of the filler due to difference in dielectric constant between the resin and the filler, which results in lowering of the dielectric breakdown voltage. The conventional film, therefore, has much room for improvement in terms of compatibility between the slidability and the voltage resistance.

(7) On the other hand, roughening film surface allows a gap to be easily formed between overlapping films in production of a film capacitor by winding the film. It is therefore presumed that roughening film surface makes the self-healing function work easily when dielectric breakdown occurs.

(8) However, in the case where the roughened portion of the film surface has a low hardness, a gap is hardly formed between the overlapping films in production of a film capacitor through winding of the film, because the roughened portion of the film is easily collapsed. As a result, the decomposition gas from the film hardly scatters from inside the film capacitor when dielectric breakdown occurs, so that the self-healing function hardly works.

(9) The present invention has been made to solve the problem, and an object thereof is to provide a film capacitor including a dielectric film excellent in slidability and voltage resistance, capable of imparting excellent pressing properties and self-healing properties. Further, another object of the present invention is to provide a film usable as dielectric film for the film capacitor. Further, another object of the present invention is to provide a metallized film usable for the film capacitor.

(10) A film capacitor of the present invention in a first aspect includes a wound body in which a dielectric film and a metal layer are wound, the dielectric film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, the dielectric film including a first main surface and a second main surface opposed to each other in a thickness direction, the metal layer being disposed at least on the first main surface of the dielectric film, wherein the first main surface of the dielectric film includes a plurality of protrusions having the second organic material; and when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface of the dielectric film, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.

(11) A film capacitor of the present invention in a second aspect includes a wound body in which a dielectric film and a metal layer are wound, the dielectric film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, the dielectric film including a first main surface and a second main surface opposed to each other in a thickness direction, the metal layer being disposed at least on the first main surface of the dielectric film, wherein the first main surface of the dielectric film includes a plurality of protrusions having the second organic material, a surface of the metal layer on an opposite side of the first main surface of the dielectric film includes a plurality of protruding parts coinciding with the plurality of the protrusions, and when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less.

(12) A film of the present invention includes a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, and includes a first main surface and a second main surface opposed to each other in a thickness direction, wherein the first main surface includes a plurality of protrusions having the second organic material, and when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or

more and less than $2.50\text{ }\mu\text{m}$ is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by $2.50\text{ }\mu\text{m}$ or more is 0.100% or less.

(13) A metallized film of the present invention includes: a film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, and including a first main surface and a second main surface opposed to each other in a thickness direction; and a metal layer disposed at least on the first main surface of the film, wherein the first main surface of the film includes a plurality of protrusions having the second organic material, a surface of the metal layer on an opposite side of the first main surface of the film includes a plurality of protruding parts coinciding with the plurality of the protrusions, and when a metal layer face positioned at an average height in the thickness direction is defined in an area range of $100\text{ }\mu\text{m}$ by $140\text{ }\mu\text{m}$ of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by $0.05\text{ }\mu\text{m}$ or more and less than $0.20\text{ }\mu\text{m}$ is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by $0.20\text{ }\mu\text{m}$ or more and less than $2.50\text{ }\mu\text{m}$ is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by $2.50\text{ }\mu\text{m}$ or more is 0.100% or less.

(14) According to the present invention, a film capacitor including a dielectric film excellent in slidability and voltage resistance, capable of imparting excellent pressing properties and self-healing properties can be provided. Further, according to the present invention, a film usable as dielectric film for the film capacitor can be provided. Further, according to the present invention, a metallized film usable for the film capacitor can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic perspective view showing an example of a film capacitor of the present invention.
- (2) FIG. 2 is a schematic cross-sectional view showing a portion corresponding to a line segment A1-A2 in FIG. 1.
- (3) FIG. 3 is a perspective schematic view showing an example of a wound body in FIG. 1 and FIG. 2.
- (4) FIG. 4 is a schematic plan view showing an example of a metal layer including a fuse part.
- (5) FIG. 5 is a schematic plan view showing an example of a film of the present invention.
- (6) FIG. 6 is a schematic cross-sectional view showing a portion corresponding to a line segment B1-B2 in FIG. 5.
- (7) FIG. 7 is a schematic plan view showing an example of a metallized film to constitute the film capacitor of the present invention.
- (8) FIG. 8 is a schematic cross-sectional view showing a portion corresponding to a line segment C1-C2 in FIG. 7.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (9) The film capacitor of the present invention, the film of the present invention, and the metallized film of the present invention will be described as follows. The present invention is not limited to the following constitution, and may be appropriately modified in a range not deviating from the gist of the present invention. Further, the present invention also includes a combination of individual preferable constitutions to be described below.
- (10) The film capacitor of the present invention in the first aspect and the second aspect includes a wound body in which a dielectric film and a metal layer are wound, the dielectric film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, the dielectric film including a first main surface and a second main surface opposed to each other in a thickness direction, the metal layer

being disposed at least on the first main surface of the dielectric film.

(11) In the case where the film capacitors of the present invention in the first aspect and the second aspect are not particularly distinguished, a term “film capacitor of the present invention” is simply used.

(12) In the following, as an example of the film capacitor of the present invention, a so-called wound-type film capacitor in which a laminate of a metallized film including a metal layer disposed at least on one main surface of a dielectric film is wound will be described.

(13) FIG. 1 is a schematic perspective view showing an example of a film capacitor of the present invention. FIG. 2 is a schematic cross-sectional view showing a portion corresponding to a line segment A1-A2 in FIG. 1. FIG. 3 is a perspective schematic view showing an example of a wound body in FIG. 1 and FIG. 2.

(14) In the present specification, as shown in FIG. 1, FIG. 2 and FIG. 3, a lamination direction and a width direction of a film capacitor are directions specified by T and W, respectively. Although it can be said that a wound-type film capacitor has a plurality of lamination directions, the direction is specified by T in the present specification. The lamination direction T and the width direction W are orthogonal to each other.

(15) As shown in FIG. 1 and FIG. 2, a film capacitor **10** includes a wound body **40**, a first external electrode **41** disposed on one end face of the wound body **40**, and a second external electrode **42** disposed on another end face of the wound body **40**. Both of the end faces of the wound body **40** are opposed to each other in the width direction W.

(16) As shown in FIG. 2 and FIG. 3, the wound body **40** is a wound body in which a first metallized film **11** and a second metallized film **12** laminated in the lamination direction T are wound. In other words, the film capacitor **10** is a wound-type film capacitor including the wound body **40**.

(17) In the film capacitor **10**, it is preferable that the wound body **40** have a flat cross section perpendicular to a scroll axis direction of the wound body **40** from the viewpoint of height reduction of the film capacitor **10**. More specifically, it is preferable that the wound body **40** be pressed to have a flat cross section such as ellipse or oval, so that the cross section of the wound body **40** has a smaller thickness than that of true circle.

(18) It can be checked whether the wound body has been pressed to have a flat cross section by, for example, presence of a press mark on the wound body.

(19) The film capacitor **10** may have a winding axis in a cylindrical shape. The winding axis is disposed on the central axis of the first metallized film **11** and the second metallized film **12** in a wound state, functioning as scroll axis when the first metallized film **11** and the second metallized film **12** are wound.

(20) The first metallized film **11** includes a first dielectric film **13** and a first metal layer **15**.

(21) The first dielectric film **13** includes a first main surface **13a** and a second main surface **13b** opposed to each other in the thickness direction (lamination direction T in FIG. 2).

(22) The first metal layer **15** is disposed on the first main surface **13a** of the first dielectric film **13**. More specifically, the first metal layer **15** is disposed to reach one side edge of the first dielectric film **13** but not to reach another side edge of the first dielectric film **13** in the width direction W.

(23) The second metallized film **12** includes a second dielectric film **14** and a second metal layer **16**.

(24) The second dielectric film **14** includes a first main surface **14a** and a second main surface **14b** opposed to each other in the thickness direction (lamination direction T in FIG. 2).

(25) The second metal layer **16** is disposed on the first main surface **14a** of the second dielectric film **14**. More specifically, the second metal layer **16** is disposed not to reach one side edge of the second dielectric film **14** but to reach another side edge of the second dielectric film **14** in the width direction W.

(26) In the wound body **40**, the neighboring first metallized film **11** and second metallized film **12**

are displaced in the width direction W, such that the end of the first metal layer **15** which reaches the side edge of the first dielectric film **13** is exposed to one end face of the wound body **40** and the end of the second metal layer **16** which reaches the side edge of the second dielectric film **14** is exposed to another end face of the wound body **40**.

(27) The wound body **40** includes the first metallized film **11** and the second metallized film **12** laminated in the lamination direction T and wound, and therefore can also be called a wound body in which the first metal layer **15**, the first dielectric film **13**, the second metal layer **16**, and the second dielectric film **14** are sequentially laminated in the lamination direction T and wound.

(28) In the wound body **40**, the first metallized film **11** and the second metallized film **12** laminated in the lamination direction T are wound, such that the first metallized film **11** is disposed inside the second metallized film **12**, the first metal layer **15** is disposed inside the first dielectric film **13**, and the second metal layer **16** is disposed inside the second dielectric film **14**. In other words, the first metal layer **15** and the second metal layer **16** are opposed to each other across the first dielectric film **13** or the second dielectric film **14**.

(29) The second metal layer **16** may be disposed on the second main surface **13b** of the first dielectric film **13**, instead of the first main surface **14a** of the second dielectric film **14**. In this case, in the wound body **40**, a metallized film including the first metal layer **15** disposed on the first main surface **13a** of the first dielectric film **13** and the second metal layer **16** disposed on the second main surface **13b** of the first dielectric film **13** and the second dielectric film **14** laminated in the lamination direction T are wound.

(30) It is preferable that each of the first metal layer **15** and the second metal layer **16** include a fuse part.

(31) FIG. **4** is a schematic plan view showing an example of a metal layer including a fuse part.

(32) As shown in FIG. **4**, the first metal layer **15** includes a plurality of split electrode parts **61**, an electrode part **62** and fuse parts **63**.

(33) The split electrode parts **61** are partitioned with insulating slits **64** and opposed to the second metal layer **16** in the wound body **40**.

(34) The electrode part **62** is adjacent to the split electrode parts **61** across the insulating slits **64**, and is not opposed to the second metal layer **16** in the wound body **40**.

(35) The fuse part **63** is a part connecting each of the split electrode parts **61** to the electrode part **62**, and is narrower than the split electrode part **61** and the electrode part **62**.

(36) The electrode pattern of the first metal layer **15** including the fuse part may be, for example, an electrode pattern disclosed in JP 2004-363431 A or JP H05-251266 A, instead of the electrode pattern shown in FIG. **4**. The same is true for the electrode pattern of the second metal layer **16** including the fuse part.

(37) The first external electrode **41** is disposed on one end face of the wound body **40** and connected to the first metal layer **15** through contact with the exposed end of the first metal layer **15**.

(38) From the viewpoint of connectivity between the first metal layer **15** and the first external electrode **41**, it is preferable that the first metallized film **11** protrude in the width direction W relative to the second metallized film **12** at one end face of the wound body **40**.

(39) The second external electrode **42** is disposed on another end face of the wound body **40** and connected to the second metal layer **16** through contact with the exposed end of the second metal layer **16**.

(40) From the viewpoint of connectivity between the second metal layer **16** and the second external electrode **42**, it is preferable that the second metallized film **12** protrude in the width direction W relative to the first metallized film **11** at another end face of the wound body **40**.

(41) Examples of the material for constituting each of the first external electrode **41** and the second external electrode **42** include a metal such as zinc, aluminum, tin, and a zinc-aluminum alloy.

(42) The first external electrode **41** and the second external electrode **42** are formed preferably by

thermally spraying the metal described above on one end face and another end face of the wound body **40**, respectively.

(43) The constitution of the wound body **40** may be different from the constitution shown in FIG. 2. For example, in the first metallized film **11**, the first metal layer **15** may be divided into two metal layers in the width direction *W* such that one metal layer reaches one side edge of the first dielectric film **13** and another metal layer reaches another side edge of the first dielectric film **13**. In this case, in the first metal layer **15**, one metal layer is connected to the first external electrode **41** and another metal layer is connected to the second external electrode **42**, while the second metal layer **16** is disposed not to be connected to either the first external electrode **41** or the second external electrode **42**. As a result, a capacitor can be formed by the first metal layer **15** and the second metal layer **16**.

(44) In the film capacitor of the present invention, the film of the present invention may be used as dielectric film.

(45) The film of the present invention includes a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, and includes a first main surface and a second main surface opposed to each other in a thickness direction.

(46) FIG. 5 is a schematic plan view showing an example of a film of the present invention. FIG. 6 is a schematic cross-sectional view showing a portion corresponding to a line segment B1-B2 in FIG. 5.

(47) As shown in FIG. 5 and FIG. 6, a film (dielectric film) **110** includes a first main surface **110a** and a second main surface **110b** opposed to each other in the thickness direction.

(48) The film **110** includes a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group. More specifically, the film **110** includes a cured product having a urethane bond obtained through a reaction between the hydroxy group (OH group) of the first organic material and the isocyanate group (NCO group) of the second organic material.

(49) The presence of the urethane bond in the film can be checked by analysis using a Fourier transform infrared spectrometer (FT-IR).

(50) It is preferable that the first organic material be a polyol having a plurality of hydroxy groups in a molecule.

(51) Examples of the polyol include a polyvinyl acetal such as polyvinyl acetoacetyl, a polyether polyol such as phenoxy resin, and a polyester polyol.

(52) As the polyol, a phenoxy resin is preferred.

(53) As the first organic material, a plurality of types of materials may be used in combination.

(54) It can also be said that the second organic material is a so-called aromatic isocyanate which is an aromatic compound having an isocyanate group.

(55) The second organic material functions as a curing agent for curing a resin solution in production of the film **110** by formation of a cross-linked structure through reaction with the hydroxy group of the first organic material.

(56) It is preferable that the second organic material be a so-called aromatic polyisocyanate which is an aromatic compound having a plurality of isocyanate groups in a molecule.

(57) Examples of the aromatic polyisocyanate include diphenylmethane diisocyanate (MDI) and tolylene diisocyanate (TDI). As the aromatic polyisocyanate, a modified product of these aromatic polyisocyanate may be used.

(58) As the aromatic polyisocyanate, MDI is preferred. As the MDI, for example, a polymeric MDI or a monomeric MDI may be used.

(59) As the second organic material, a plurality types of materials may be used in combination.

(60) In the film capacitor **10** shown in FIG. 2, the film **110** may be used for both of the first dielectric film **13** and the second dielectric film **14**, or may be used for any one of the first

dielectric film **13** and the second dielectric film **14**. In the case where the film **110** is used for both of the first dielectric film **13** and the second dielectric film **14**, the compositions of the first dielectric film **13** and the second dielectric film **14** may be different from each other, though it is preferable that the compositions be the same.

(61) The film **110** is produced by applying a resin solution containing the first organic material and the second organic material to the surface of a substrate, drying the applied solution, and then heating the applied solution to be cured. The resulting film **110** is peeled off from the substrate for use.

(62) In the film of the present invention, the first main surface includes a plurality of protrusions having the second organic material.

(63) In the film **110** shown in FIG. 5 and FIG. 6, the first main surface **110a** includes a plurality of protrusions **120**. Further, the first main surface **110a** of the film **110** includes a flat part **130** including no protrusion **120**.

(64) The presence of the protrusions can be identified as black parts through observation of the first main surface of the film using a scanning electron microscope (SEM).

(65) The protrusion **120** has the second organic material, i.e., an aromatic isocyanate. More specifically, the aromatic isocyanate which the protrusion **120** has derives from an aromatic isocyanate constituting the cured product which the film **110** includes.

(66) Having an aromatic isocyanate, the protrusion **120** has a high hardness resulting from the aromatic ring of the aromatic isocyanate. Accordingly, in production of a film capacitor from the film **110**, the protrusions **120** are hardly collapsed by winding of the film **110** or by pressing after winding, so that a gap is easily formed between the overlapping films **110**. As a result, in dielectric breakdown, a decomposition gas from the film **110** easily scatters from inside the film capacitor, so that the film capacitor has excellent self-healing properties.

(67) In contrast, in the case where the protrusion **120** has an aliphatic isocyanate, the protrusion **120** has a lower hardness in comparison with a case where the protrusion **120** has an aromatic isocyanate. Accordingly, in production of a film capacitor from the film **110**, the protrusions **120** are easily collapsed by winding of the film **110** or by pressing after winding, so that a gap is hardly formed between the overlapping films **110**. As a result, in dielectric breakdown, a decomposition gas from the film **110** hardly scatters from inside the film capacitor, so that the film capacitor has lowered self-healing properties.

(68) Even with the protrusion **120** having an aromatic isocyanate, a gap is hardly formed between the overlapping films **110** in production of a film capacitor by winding the film **110**, for example, when, in the first main surface **110a** of the film **110**, an area ratio of a region having a height higher than the film face to be described later by 0.20 μm or more and less than 2.50 μm is less than 0.0998%, so that the film capacitor has lowered self-healing properties.

(69) The presence of the aromatic isocyanate in the protrusion can be checked as described below. First, using a Fourier transform infrared spectrometer (FT-IR) "FT/IR-4100ST" manufactured by JASCO Corporation, an infrared absorption spectrum of the protrusion in a measurement wavenumber range of 500 $\text{cm}.\text{sup}.-1$ to 4000 $\text{cm}.\text{sup}.-1$ is measured by attenuated total reflection method (ATR). In the infrared absorption spectrum of the protrusion, confirmed detection of the absorption peak of an aromatic ring and the absorption peak of an isocyanate group allows to confirm the presence of an aromatic isocyanate in the protrusion. For example, in the case of using MDI as the aromatic isocyanate, the absorption peak of the aromatic ring is observed in a measurement wavenumber range of 1450 $\text{cm}.\text{sup}.-1$ to 1550 $\text{cm}.\text{sup}.-1$, and the absorption peak of the isocyanate group is observed in a measurement wavenumber range of 2200 $\text{cm}.\text{sup}.-1$ to 2400 $\text{cm}.\text{sup}.-1$ in the infrared absorption spectrum.

(70) Further, by the same method, in the infrared absorption spectrum of the flat part, confirmed detection of the absorption peak of an aromatic ring and the absorption peak of an isocyanate group allows to confirm that the aromatic isocyanate which the protrusion has derives from the aromatic

isocyanate constituting the cured product which the film includes.

(71) In the film of the present invention, when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.

(72) In the film of the present invention, the area ratio of the region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm may be 0.00% or more. Also, the area ratio of the region having a height higher than the film face by 2.50 μm or more may be 0.000% or more.

(73) In the film **110** shown in FIG. 5 and FIG. 6, when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface **110a**, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.

(74) In order to enhance the slidability for sliding the films **110** against each other, it is important to reduce the contact area between the films **110**. From such a viewpoint, the first main surface **110a** of the film **110** includes the plurality of the protrusions **120** so as to reduce the contact area between the films **110** when the films **110** are slid against each other. However, depending on the height distribution of the first main surface **110a** of the film **110** including protrusions **120**, the compatibility between slidability and voltage resistance may not be achieved. In contrast, regarding the height distribution of the first main surface **110a** of the film **110**, when the film face positioned at an average height in the thickness direction is defined in the area range describe above, the area ratio of the region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm being 6.04% or less, the area ratio of the region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm being 0.0998% to 1.13%, and the area ratio of the region having a height higher than the film face by 2.50 μm or more being 0.100% or less allows the film **110** to have excellent slidability and voltage resistance. Further, due to having excellent slidability, the film **110** is capable of imparting excellent pressing properties and self-healing properties when wound for constitution of a film capacitor. In the case where the height distribution of the first main surface **110a** of the film **110** is in the range described above, it can be said that a plurality of protrusions **120** having an appropriate height are distributed in the first main surface **110a** at an appropriate ratio.

(75) When, in the first main surface **110a** of the film **110**, the area ratio of the region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is more than 6.04%, many high protrusions **120** on which an electrical field tends to be concentrated are present, so that the voltage resistance is lowered.

(76) When, in the first main surface **110a** of the film **110**, the area ratio of the region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is less than 0.0998%, the contact area between the flat parts **130** tends to be large in sliding the films **110** against each other, so that the slidability is lowered. When, in the first main surface **110a** of the film **110**, the area ratio of the region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is more than 1.13%, many high protrusions **120** on which an electrical field tends to be concentrated are present, so that the voltage resistance is lowered.

(77) When, in the first main surface **110a** of the film **110**, the area ratio of the region having a height higher than the film face by 2.50 μm or more is more than 0.100%, many high protrusions **120** on which an electrical field tends to be concentrated are present, so that the voltage resistance is lowered.

(78) Regarding the height distribution of the first main surface **110a** of the film **110**, when the film

face positioned at an average height in the thickness direction is defined in the area range described above, and when the area ratio of the region having a height higher than the film face by $0.05\text{ }\mu\text{m}$ or more and less than $0.20\text{ }\mu\text{m}$ is 6.04% or less, the area ratio of the region having a height higher than the film face by $0.20\text{ }\mu\text{m}$ or more and less than $2.50\text{ }\mu\text{m}$ is 0.0998% to 1.13%, and the area ratio of the region having a height higher than the film face by $2.50\text{ }\mu\text{m}$ or more is 0.100% or less, the first main surface **110a** of the film **110** may include a plurality of concaves in addition to the plurality of the protrusions **120**.

(79) The height distribution of the first main surface of the film is determined as follows.

(80) First, using a laser microscope “VK-8700” manufactured by Keyence Corporation, the first main surface of the film is enlarged by 100 times to observe an area range of $100\text{ }\mu\text{m}$ by $140\text{ }\mu\text{m}$. On this occasion, aluminum may be vapor deposited to a thickness of 10 nm on the first main surface of the film in advance. In the area range described above, a direction along a length of $100\text{ }\mu\text{m}$ is defined as a first direction and a direction along a length of $140\text{ }\mu\text{m}$ is defined as a second direction in the following.

(81) Subsequently, the area range described above is divided into 786432 cells in total, i.e., 1024 cells in the first direction by 768 cells in the second direction. The height data in the thickness direction (average of height in a cell) is acquired for each of the 786432 cells divided using an analysis software “VK-analyzer” dedicated to the laser microscope “VK-8700” manufactured by Keyence Corporation. Then the average of the height data of 786432 cells is calculated, and the resulting average is defined as the average height of the film in the thickness direction. Also, a virtual face located at the average height in the thickness direction is defined as the film face.

(82) Then, the number of cells having height data higher than the film face by $0.05\text{ }\mu\text{m}$ or more, i.e., J1, is counted to calculate the area ratio of cells having height data higher than the film face by $0.05\text{ }\mu\text{m}$ or more, i.e., K1, as “ $100\times J1/786432$ ”. Also, the number of cells having height data higher than the film face by $0.20\text{ }\mu\text{m}$ or more, i.e., J2, is counted to calculate the area ratio of cells having height data higher than the film face by $0.20\text{ }\mu\text{m}$ or more, i.e., K2, as “ $100\times J2/786432$ ”. Also, the number of cells having height data higher than the film face by $2.50\text{ }\mu\text{m}$ or more, i.e., J3, is counted to calculate the area ratio of cells having height data higher than the film face by $2.50\text{ }\mu\text{m}$ or more, i.e., K3, as “ $100\times J3/786432$ ”.

(83) Then, based on each of the resulting area ratios, the area ratio of the region having a height higher than the film face by $0.05\text{ }\mu\text{m}$ or more and less than $0.20\text{ }\mu\text{m}$ is calculated as “K1-K2”. Also, the area ratio of the region having a height higher than the film face by $0.20\text{ }\mu\text{m}$ or more and less than $2.50\text{ }\mu\text{m}$ is calculated as “K2-K3”. Further, the area ratio of the region having a height higher than the film face by $2.50\text{ }\mu\text{m}$ or more is defined as “K3”. The height distribution of the first main surface of the film is thus determined.

(84) The planar shape of the protrusion **120** may be in a circular form as shown in FIG. 5, an ellipse form, or other forms.

(85) The planar shapes of the protrusions **120** may be the same as or different from each other.

(86) The cross-sectional shape of the protrusion **120** may be in a tapered form as shown in FIG. 6, or may be in a form other than the tapered form.

(87) The cross-sectional shapes of the protrusions **120** may be the same as or different from each other.

(88) It is preferable that the top face of the protrusion **120** be dented as shown in FIG. 6. In this case, the contact area between the protrusions **120** tends to be small when the films **110** are slid against each other, so that the slidability tends to be enhanced. The top face of the protrusion **120** needs not to be dented.

(89) The tip of the protrusion **120** may be in a pointed form or in a round form.

(90) The second main surface **110b** of the film **110** may include no protrusion or may include a plurality of protrusions. In this case, regarding the height distribution of the second main surface **110b** of the film **110**, when a film face positioned at an average height in the thickness direction is

defined in an area range of $100\text{ }\mu\text{m}$ by $140\text{ }\mu\text{m}$ of the second main surface **110b**, it is preferable that an area ratio of a region having a height higher than the film face by $0.05\text{ }\mu\text{m}$ or more and less than $0.20\text{ }\mu\text{m}$ be 6.04% or less, an area ratio of a region having a height higher than the film face by $0.20\text{ }\mu\text{m}$ or more and less than $2.50\text{ }\mu\text{m}$ be 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by $2.50\text{ }\mu\text{m}$ or more be 0.100% or less. On this occasion, the second main surface **110b** of the film **110** may include a plurality of concaves in addition to the plurality of the protrusions.

(91) The height distribution of the second main surface of the film is determined in the same manner as in the height distribution of the first main surface of the film, except that the observation target is the second main surface of the film.

(92) It is preferable that, in the film of the present invention, a coefficient of static friction on the first main surface side be 1.0 or less.

(93) It is preferable that, in the film **110** shown in FIG. 5 and FIG. 6, a coefficient of static friction on the first main surface **110a** side be 1.0 or less. In this case, the slidability of the film **110** is exceptionally good.

(94) On the other hand, with a too small coefficient of static friction on the first main surface **110a** side of the film **110**, winding deviation of the film **110** towards width direction occurs during winding of the film **110** for production of a film capacitor, so that an external electrode may be hardly formed on the end face of the resulting wound body in a later step. From such a viewpoint, the coefficient of static friction on the first main surface **110a** side of the film **110** is preferably 0.1 or more.

(95) The coefficient of static friction of the film is determined as follows. First, two sheets of films are prepared as measurement specimens. Regarding both of the main surfaces of the measurement specimens, the main surface on the substrate side during production is defined as mold-release surface, and the main surface on the opposite side of the substrate is defined as drying surface. More specifically, regarding each of the measurement specimens, the drying surface corresponds to a first main surface and the mold-release surface corresponds to a second main surface. Further, it is preferable that the length direction of each of the measurement specimens be the same as the direction in which a tensile stress is applied to the film in the production step of a film capacitor, for example, the winding direction of the film. Subsequently, on the mold-release surface of one of the measurement specimens, a rectangular plate is fixed such that the drying surface is exposed. Further, on the drying surface of another one of the measurement specimens, a rectangular weight with a weight of 200 g is attached such that the mold-release surface is exposed. Then, the mold-release surface of the other measurement specimen with the drying surface to which the weight is attached comes into contact with the drying surface of the one measurement specimen with the mold-release surface to which the plate is fixed, such that the length directions thereof are parallel to each other. The weight attached to the drying surface of the other measurement specimen is pulled at a rate of 150 mm/minute in the length direction after fixation to a force gauge manufactured by Imada Inc. On this occasion, the maximum frictional force until occurrence of motion of the weight together with the other measurement specimen is read, and the static frictional coefficient is calculated from the read-out value.

(96) It is preferable that the film **110** have a glass transition point of 130°C. or more. In this case, the film **110** has excellent heat resistance, so that the guaranteed temperature of a film capacitor made of the film **110** may be raised to, for example, 125°C. or more.

(97) The glass transition point of the film is determined as follows. First, using a dynamic viscoelasticity measuring (DMA) apparatus “RSA-III” manufactured by TA Instruments Inc., the storage modulus and loss modulus of the film are measured while raising temperature from room temperature to 250°C. at a temperature rising rate of $10^{\circ}\text{C./minute}$ under measurement conditions at a measurement frequency of 10 rad/second, and a strain of 0.1%. Temperature at which the loss tangent ($\tan\delta$) represented by loss modulus/storage modulus has a maximum peak value is defined

as the glass transition point.

(98) The thickness **S** of the film **110** is preferably 1 μm to 10 μm , more preferably 3 μm to 5 μm .

(99) The thickness **S** of the film is a thickness determined at a position where the protrusion **120** is absent, as shown in FIG. 6.

(100) The thickness of the film can be measured by using an optical film thickness meter.

(101) The film of the present invention is made into a metallized film by providing at least the first main surface with a metal layer, so that the film capacitor of the present invention can be constituted.

(102) FIG. 7 is a schematic plan view showing an example of a metallized film to constitute the film capacitor of the present invention. FIG. 8 is a schematic cross-sectional view showing a portion corresponding to a line segment C1-C2 in FIG. 7.

(103) As shown in FIG. 7 and FIG. 8, a metallized film **210** includes the film **110** shown in FIG. 5 and FIG. 6 and a metal layer **220** disposed on the first main surface **110a** of the film **110**.

(104) In the film capacitor of the present invention, the first main surface of the dielectric film includes a plurality of protrusions having the second organic material.

(105) In the film **110** shown in FIG. 8, in the same manner as in the film **110** shown in FIG. 5 and FIG. 6, the first main surface **110a** includes a plurality of protrusions **120** having the second organic material, i.e., an aromatic isocyanate. Also, the first main surface **110a** of the film **110** includes a flat part **130** including no protrusion **120**.

(106) Since the protrusion **120** has an aromatic isocyanate, the film capacitor made of the metallized film **210** including the metal layer **220** disposed on the first main surface **110a** of the film **110** has excellent self-healing properties.

(107) In the first aspect of the film capacitor of the present invention, when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface of the dielectric film, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less. As described above, in the first aspect of the film capacitor of the present invention, attention is focused on the height distribution of the first main surface of the dielectric film including the plurality of the protrusions.

(108) In the first aspect of the film capacitor of the present invention, the area ratio of the region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm may be 0.00% or more. Also, the area ratio of the region having a height higher than the film face by 2.50 μm or more may be 0.000% or more.

(109) In the film **110** shown in FIG. 8, as in the film **110** shown in FIG. 5 and FIG. 6, when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface **110a**, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.

(110) In the area range described above of the first main surface **110a** of the film **110**, when the film face positioned at an average height in the thickness direction is defined, the area ratio of the region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm being 6.04% or less, the area ratio of the region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm being 0.0998% to 1.13%, and the area ratio of the region having a height higher than the film face by 2.50 μm or more being 0.100% or less allows the film **110** to have excellent slidability and voltage resistance. Further, due to having excellent slidability, the film **110** is capable of imparting excellent pressing properties and self-healing properties when

wound for constitution of a film capacitor.

(111) In measurement of the height distribution of the first main surface of the film in the film capacitor, the measurement is performed by the method described above for a region including no metal layer in the metallized film positioned at the outermost surface of the film capacitor. On this occasion, in the case there exists a masking oil attached to the surface in the measurement region, it is preferable that the measurement be performed after removal of the masking oil with a solvent such as hexane and toluene.

(112) In the film capacitor of the present invention, the metallized film of the present invention may be used as metallized film.

(113) The metallized film of the present invention includes: a film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, and including a first main surface and a second main surface opposed to each other in a thickness direction; and a metal layer disposed at least on the first main surface of the film. Also, in the metallized film of the present invention, the first main surface of the film includes a plurality of protrusions having the second organic material. Further, in the metallized film of the present invention, a surface of the metal layer on an opposite side of the first main surface of the film includes a plurality of protruding parts coinciding with the plurality of the protrusions.

(114) In the first aspect of the film capacitor of the present invention, it is preferable that a surface of the metal layer on an opposite side of the first main surface of the dielectric film includes a plurality of protruding parts coinciding with the plurality of the protrusions, in the same manner as in the second aspect of the film capacitor of the present invention to be described later.

(115) In the first aspect of the film capacitor of the present invention, in the case where the surface of the metal layer includes the plurality of the protruding parts, it is preferable that when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm be 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm be 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more be 0.100% or less in the same manner as in the second aspect of the film capacitor of the present invention to be described later.

(116) In the second aspect of the film capacitor of the present invention, a surface of the metal layer on an opposite side of the first main surface of the dielectric film includes a plurality of protruding parts coinciding with the plurality of the protrusions.

(117) In the metallized film **210** shown in FIG. 7 and FIG. 8, the metal layer **220** includes a surface **220a** on the opposite side of the first main surface **110a** of the film **110**, with a plurality of protruding parts **230** coinciding with the protrusions **120**. Further, the surface **220a** of the metal layer **220** includes a flat part **240** including no protruding part **230**.

(118) The presence of the protruding part can be identified as a black part through observation of the surface of the metal layer using a scanning electron microscope.

(119) In the metallized film of the present invention, when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less.

(120) In the metallized film of the present invention, the area ratio of the region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm may be 0.00% or more. Also, the area ratio of the region having a height higher than the metal layer face by 2.50 μm or

more may be 0.000% or more.

(121) In the second aspect of the film capacitor of the present invention, when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less. As described above, in the second aspect of the film capacitor of the present invention, attention is focused on the height distribution of the surface of the metal layer including the plurality of the protruding parts.

(122) In the second aspect of the film capacitor of the present invention, the area ratio of the region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm may be 0.00% or more. Also, the area ratio of the region having a height higher than the metal layer face by 2.50 μm or more may be 0.000% or more.

(123) In the metallized film **210** shown in FIG. 7 and FIG. 8, when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface **220a** of the metal layer **220**, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less.

(124) When the metal layer face positioned at an average height in the thickness direction is defined in the area range of the surface **220a** of the metal layer **220**, the area ratio of the region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm being 6.17% or less, the area ratio of the region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm being 0.118% to 1.24%, and the area ratio of the region having a height higher than the metal layer face by 2.50 μm or more being 0.100% or less allows the metallized film **210** to have excellent slidability and voltage resistance. Further, due to having excellent slidability, the metallized film **210** is capable of imparting excellent pressing properties and self-healing properties when wound for constitution of a film capacitor. In the case where the height distribution of the surface **220a** of the metal layer **220** is in the range described above, it can be said that a plurality of protruding parts **230** having an appropriate height are distributed in the surface **220a** at an appropriate ratio.

(125) When, in the surface **220a** of the metal layer **220**, the area ratio of the region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is more than 6.17%, many high protruding parts **230** on which an electrical field tends to be concentrated are present, so that the voltage resistance is lowered.

(126) When, in the surface **220a** of the metal layer **220**, the area ratio of the region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is less than 0.118%, the contact area between the flat parts **240** tends to be large in sliding the metallized films **210** against each other, so that the slidability is lowered. When, in the surface **220a** of the metal layer **220**, the area ratio of the region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is more than 1.24%, many high protrusions **230** on which an electrical field tends to be concentrated are present, so that the voltage resistance is lowered.

(127) When, in the surface **220a** of the metal layer **220**, the area ratio of the region having a height higher than the metal layer face by 2.50 μm or more is more than 0.100%, many high protruding parts **230** on which an electrical field tends to be concentrated are present, so that the voltage resistance is lowered.

(128) Regarding the height distribution of the surface **220a** of the metal layer **220**, when the metal layer face positioned at an average height in the thickness direction is defined in the area range

described above, and when the area ratio of the region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, the area ratio of the region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and the area ratio of the region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less, the surface **220a** of the metal layer **220** may include a plurality of concaves in addition to the plurality of the protrusions **230**.

(129) The height distribution of the surface of the metal layer is determined in the same manner as the height distribution of the first main surface of the film, except that the observation target is the surface of the metal layer.

(130) In measurement of the height distribution of the surface of the metal layer of a film capacitor, the measurement is performed in a specified region of the metallized film positioned at the outermost surface of the film capacitor.

(131) In the case of the wound body **40** shown in FIG. 3, it is preferable that the measurement of the height distribution of the surface of the second metal layer **16** be performed in a region L of the second metallized film **12** positioned at the outermost surface provided with the second metal layer **16**, by the method described above. The region L of the second metallized film **12** is in a square shape having a length M both in the width direction W and the length direction (winding direction). Further, when a centerline Q passing through a center point P of an end side N of the second metallized film **12** and extending in the length direction is defined, a region L of the second metallized film **12** is linearly symmetrical with respect to the centerline Q in the width direction W. The length M of the region L of the second metallized film **12** is 10% of the length of the end side N of the second metallized film **12**.

(132) In the metallized film of the present invention, it is preferable that a coefficient of static friction on a side of the metal layer on the first main surface of the film be 1.4 or less.

(133) In the first aspect of the film capacitor of the present invention, in the case where the surface of the metal layer includes the plurality of the protruding parts, it is preferable that a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film be 1.4 or less.

(134) In the second aspect of the film capacitor of the present invention, it is preferable that a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film be 1.4 or less.

(135) In the metallized film **210** shown in FIG. 7 and FIG. 8, it is preferable that a coefficient of static friction on the surface **220a** side of the metal layer **220** be 1.4 or less. In this case, the slidability of the metallized film **210** is exceptionally good.

(136) On the other hand, with a too small coefficient of static friction on the surface **220a** side of the metal layer **220** in the metallized film **210**, winding deviation of the metallized film **210** towards width direction occurs during winding of the metallized film **210** for production of a film capacitor, so that an external electrode may be hardly formed on the end face of the resulting wound body in a later step. From such a viewpoint, the coefficient of static friction on the surface **220a** side of the metal layer **220** in the metallized film **210** is preferably 0.2 or more.

(137) The coefficient of static friction of the metallized film is determined in the same manner as the coefficient of static friction of the film, except that a metallized film is used as measurement specimen.

(138) Examples of the material to constitute the metal layer **220** include a metal such as aluminum, zinc, titanium, magnesium, tin and nickel.

(139) The thickness of the metal layer **220** is preferably 5 nm to 40 nm.

(140) The thickness of the metal layer can be specified through observation of the cross section of the metallized film in the thickness direction using a transmission electron microscope (TEM).

(141) The film of the present invention may be produced, for example, by the following method.

(142) Step of Preparing Resin Solution

(143) A resin solution is prepared by mixing a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group.

(144) As the first organic material and the second organic material, ones described above are used.

(145) In preparation of the resin solution, the first organic material and the second organic material may be diluted with a solvent. It is particularly preferable that the first organic material and the second organic material be diluted with a mixed solvent containing a first solvent selected from ketones and a second solvent selected from cyclic ether compounds.

(146) Examples of the ketones from which the first solvent is selected include methyl ethyl ketone and diethyl ketone.

(147) As the first solvent, a plurality of types of ketones may be used in combination.

(148) Examples of the cyclic ether compounds from which the second solvent is selected include tetrahydrofuran and tetrahydropyran.

(149) As the second solvent, a plurality of types of cyclic ether compounds may be used in combination.

(150) It is preferable that as the solvent, a mixed solvent containing methyl ethyl ketone and tetrahydrofuran be used.

(151) Drying and Curing Step of Resin Solution

(152) First, the resin solution is applied to the surface of a substrate.

(153) Examples of the substrate include a polyethylene terephthalate film and a polypropylene film.

(154) Subsequently, the resulting coating film of the resin solution is dried in a drying furnace, and then cured through a heating treatment. A film is thus made on the surface of the substrate.

(155) On this occasion, the coating film of the resin solution in the drying furnace is subject to the adjustment of drying temperature, drying time, air flow, etc., so that the second organic material, i.e., aromatic isocyanate, agglomerates on the drying surface as main surface of the coating film on the opposite side of the substrate. As a result, the agglomerates cause a plurality of protrusions, of which height, etc. are also subject to control. Thus, regarding the film state resulting from curing the coating film, the height distribution of the first main surface of the film can be controlled such that when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface corresponding to the drying surface, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.

(156) It is preferable that the drying temperature of the coating film be adjusted in the range of 70° C. to 150° C.

(157) The drying time of the coating film is adjustable through the transportation rate of the substrate with the coating film transported in a drying furnace. It is preferable that transportation rate be adjusted in the range of 100 m/minute to 160 m/minute.

(158) The resulting film is peeled off from the substrate for use. As described above, the first main surface of the film corresponds to the drying surface as main surface of the coating film on the opposite side of the substrate. Also, the second main surface of the film corresponds to the mold-release surface of the coating film on the substrate side.

(159) As described above, the film shown in FIG. 5 and FIG. 6 is prepared.

(160) The metallized film of the present invention is produced, for example, by the following method.

(161) Step of Preparing Metallized Film

(162) First, as the first dielectric film and the second dielectric film, the film shown in FIG. 5 and FIG. 6 is prepared by the production method of the film of the present invention as described above.

(163) Next, a metal is vapor deposited on the first main surface of the first dielectric film to form a

first metal layer, so that a first metallized film is prepared. On this occasion, the first metal layer is formed such that the surface of the first metal layer on the opposite side of the first main surface of the first dielectric film includes a plurality of protruding parts coinciding with the protrusions of the first dielectric film. Further, in the width direction, the first metal layer is formed to reach one side end of the first dielectric film but not to reach another side end of the first dielectric film.

(164) Also, a metal is vapor deposited on the first main surface of the second dielectric film to form a second metal layer, so that a second metallized film is prepared. On this occasion, the second metal layer is formed such that the surface of the second metal layer on the opposite side of the first main surface of the second dielectric film includes a plurality of protruding parts coinciding with the protrusions of the second dielectric film. Further, in the width direction, the second metal layer is formed not to reach one side end of the second dielectric film but to reach another side end of the second dielectric film.

(165) By the present step, the metallized films shown in FIG. 7 and FIG. 8 are prepared as the first metallized film and the second metallized film.

(166) The film capacitor of the present invention is produced, for example, by the following method.

(167) Step of Preparing Wound Body

(168) First, by the production method of the metallized film of the present invention described above, the metallized films shown in FIG. 7 and FIG. 8 are prepared as the first metallized film and the second metallized film.

(169) Next, the first metallized film and the second metallized film are overlapped with a specified displacement distance in the width direction, and then wound to prepare a wound body. On an as needed basis, the resulting wound body may be interposed in the direction vertical to the width direction to be pressed into an elliptical cylindrical shape.

(170) Step of Forming External Electrode

(171) The first external electrode is formed to make a connection to the first metal layer by thermally spraying a metal on one end face of the wound body.

(172) Also, the second external electrode is formed to make a connection to the second metal layer by thermally spraying a metal on another end surface of the wound body.

(173) As described above, the film capacitor shown in FIG. 1 and in FIG. 2 is produced.

(174) The film capacitor of the present invention is applicable to known uses, and appropriately used for power electronic equipment such as a motor-driven compressor/pump, a charger, a DC-DC converter, and a drive inverter, because the service life of the equipment used under an environment with large temperature changes at high temperature can be prolonged.

EXAMPLES

(175) Examples are shown as follows to more specifically disclose the film capacitor of the present invention, the film of the present invention, and the metallized film of the present invention. The present invention is not limited to these Examples only.

(176) Film specimens 1 to 7 were produced by the following method.

(177) Step of Preparing Resin Solution

(178) A phenoxy resin and MDI were diluted and mixed with a mixed solvent of methyl ethyl ketone and tetrahydrofuran to prepare a resin solution.

(179) Drying and Curing Step of Resin Solution

(180) First, the resin solution was applied to the surface of a polyethylene terephthalate film with a gravure coater.

(181) Next, the resulting coating film of the resin solution was dried in a drying furnace, and then cured through a heating treatment for a specified time. Thereby, film specimens 1 to 7 having a thickness of 4.5 μm were prepared on the surface of the polyethylene terephthalate film.

(182) On this occasion, the coating film of the resin solution was conditioned at a drying temperature in the range of 70° C. to 150° C., and at a transportation rate in the drying furnace in

the range of 100 m/minute to 160 m/minute to control the formation state of the protrusions having MDI, the height of the protrusions, etc., in the first main surface of each of the film specimens. Thereby, the height distribution of the first main surface of the film specimen was made different among the film specimens 1 to 7.

(183) The resulting film specimens 1 to 7 were then peeled off from the polyethylene terephthalate film.

(184) Also, in the resin solution for use in production of the film specimens 1 to 7, hexamethylene diisocyanate (HDI) which is an aliphatic polyisocyanate was blended instead of MDI to produce a film specimen 8 in the same manner as in the film specimens 1 to 7.

(185) Also, using the same resin solution for production of the film specimens 1 to 7, a film specimen 9 including no protrusion in the first main surface was produced. Further, a filler was further blended in the resin solution for use in production of the film specimen 9, and the blending ratio of the filler was changed to produce film specimens 10 to 12. The blending ratios of the filler relative to the total weight of the phenoxy resin and MDI were 1 wt % in the film specimen 10, 3 wt % in the film specimen 11, and 10 wt % in the film specimen 12, respectively. As the filler, acrylic beads “MP-1451” (average particle size: 0.15 μm) manufactured by Soken Chemical & Engineering Co., Ltd. Was used.

Evaluation

(186) It was confirmed that the protrusions of the film specimens 1 to 7 had MDI by the method described above. It was confirmed that the protrusions of the film specimen 8 had HDI. It was confirmed that the first main surface of the film specimen 9 included no protrusion. It was confirmed that the first main surface of the film specimens 10 to 12 were roughened with the filler.

(187) Further, the film specimens 1 to 12 were subjected to the following evaluation. The results are shown in Table 1. The film specimens are simply written as “Specimen” in Table 1.

(188) Height Distribution

(189) As the height distribution of the first main surface of each of the film specimens 1 to 12, the area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm , the area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm , and the area ratio of a region having a height higher than the film face by 2.50 μm or more were measured by the method described above. In Table 1, the area ratio of the region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is written as “Area ratio 1”, the area ratio of the region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is written as “Area ratio 2”, and the area ratio of the region having a height higher than the film face by 2.50 μm or more is written as “Area ratio 3”.

(190) Also, aluminum was vapor deposited on the first main surface of each of the film specimens 1 to 12 to form a metal layer, so that a metallized film was prepared. As the height distribution of the surface of the metal layer of the resulting metallized film, the area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm , the area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm , and the area ratio of a region having a height higher than the metal layer face by 2.50 μm or more were measured by the method described above. In Table 1, the area ratio of the region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is written as “Area ratio 1’”, the area ratio of the region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is written as “Area ratio 2’”, and the area ratio of the region having a height higher than the metal layer face by 2.50 μm or more is written as “Area ratio 3’”.

(191) Glass Transition Point

(192) The glass transition point of each of the film specimens 1 to 8 were measured by the method described above.

(193) Coefficient of Static Friction 1

(194) The coefficient of static friction of each of the film specimens 1 to 12 was measured by the

method described above. In Table 1, the coefficient of static friction of the specimen in the film state is written as “Coefficient of static friction 1”.

(195) Coefficient of Static Friction 2

(196) Aluminum was vapor deposited on the first main surface of the first main surface of each of the film specimens 1 to 12 to form a metal layer, so that a metallized film was prepared. The coefficient of static friction of the resulting metallized film was measured by the method described above. In Table 1, the coefficient of static friction of the specimen in the metallized film state is written as “Coefficient of static friction 2”.

(197) Dielectric Breakdown Voltage

(198) First, a metal layer was formed on both main surfaces of each film specimens 1 to 12 by vapor deposition of aluminum to prepare a measurement specimen. On this occasion, the area of the region where the deposited metal layers on both of the main surfaces of film specimen overlapping to each other was set to 3 cm.sup.2. For each of the film specimens 1 to 12, 16 each of the measurement specimens were prepared. Next, each of the electric field strengths was heled for 10 minutes with an increment of 25 V/μm for the 16 measurement specimens, and when 8 breakdown traces were generated in the film specimens, the electric field at that time was defined as failure voltage. The measurement temperature was set to 125° C. The failure voltages of the 16 measurement specimens were Weibull-plotted, and the value at which the failure frequency reached 50% in the Weibull distribution was defined as the dielectric breakdown voltage of the film specimen.

(199) Pressing Property

(200) First, aluminum was vapor deposited on the first main surface of each of the film specimens 1 to 12 to form a metal layer, so that a metallized film was prepared. On this occasion, a fluorine-based oil was applied to the first main surface of the film specimen in advance in order to draw a pattern on the metal layer. Next, the metallized film was cut to a specified width, and then the metallized film in a specified length was wound into a cylindrical shape to prepare a wound body. The wound body was then pressed with a pressure appropriately adjusted in the range of 10 N to 100 N depending on the size of the wound body, and then the pressing uniformity of the wound body was evaluated. As evaluation indexes, when the wound body was viewed from the width direction (refer to FIG. 3), the case where the film has no wrinkle/fold along the inside of the void at the center of the winding was evaluated as good, and the case where the film has wrinkles/folds along the inside of the void at the center of the winding was evaluated as poor.

(201) Self-Healing Property

(202) First, aluminum was vapor deposited on the first main surface of each of the film specimens 1 to 12 to form a metal layer, so that a metallized film was prepared. On this occasion, a fluorine-based oil was applied to the first main surface of the film specimen in advance in order to draw a pattern on the metal layer. Next, the metallized film was cut to a specified width and then the metallized film in a specified length was wound into a cylindrical shape to prepare a wound body. The external electrodes were formed by thermally spraying a metal on both end faces of the resulting wound body to produce a film capacitor. Then, while increasing the voltage applied to the resulting film capacitor gradually, evaluation whether the original applied voltage was recovered even after instantaneous decrease in the applied voltage due to dielectric breakdown was performed. As evaluation indexes, the case where the original applied voltage was recovered was evaluated as good, and the case where the original applied voltage was not recovered was evaluated as poor.

(203) TABLE-US-00001 TABLE 1 Glass Dielectric Area Area Area Area Area Area transition
Coefficient Coefficient breakdown Self- ratio 1 ratio 2 ratio 3 ratio 1' ratio 2' ratio 3' point of static
of static voltage Pressing healing (%) (%) (%) (%) (%) (%) (%) (° C.) friction 1 friction 2 (V/μm)
property property Specimen 1 6.04 0.0998 0.000 6.17 0.118 0.000 173 0.7 1.2 333 Good Good
Specimen 2 3.23 0.528 0.000 3.32 0.562 0.000 176 0.4 0.8 328 Good Good Specimen 3 0.211 1.13

0.000 0.252 1.24 0.000 178 0.3 0.8 347 Good Good *Specimen 4 0.123 0.0200 0.000 0.141 0.0400
0.000 171 1.2 1.6 339 Poor Poor *Specimen 5 0.313 1.49 0.000 0.332 1.64 0.000 172 0.3 0.7 236
Good Good *Specimen 6 8.36 0.421 0.000 8.62 0.501 0.000 173 0.6 1.0 258 Good Good
*Specimen 7 0.212 0.313 0.300 0.232 0.389 0.343 175 0.4 0.7 270 Good Good *Specimen 8 0.101
0.512 0.000 0.132 0.622 0.000 103 0.8 1.2 339 Good Poor *Specimen 9 0.100 0.0101 0.000 0.121
0.0203 0.000 — 7.3 9.0 385 Poor Poor *Specimen 10 1.22 0.0623 0.000 1.26 0.0711 0.000 — 1.1
1.5 290 Poor Poor *Specimen 11 2.63 0.112 0.000 2.69 0.122 0.000 — 1.0 1.4 254 Good Good
*Specimen 12 3.94 0.141 0.0112 4.02 0.172 0.0123 — 0.9 1.3 240 Good Good

(204) In Table 1, the specimen names marked with * represent comparative examples which are out of the scope of the present invention.

(205) Thereby, in the film specimens 1 to 3, it was confirmed that the first main surface included protrusions having an aromatic isocyanate, and it was further confirmed that the area ratio 1 was 6.04% or less, the area ratio 2 was 0.0998% to 1.13%, and the area ratio 3 was 0.100% or less. Also, in the metallized film state formed by vapor depositing a metal layer on the first main surface of the film specimens 1 to 3, it was confirmed that area ratio 1' was 6.17% or less, the area ratio 2' was 0.118% to 1.24%, and the area ratio 3' was 0.100% or less. The film specimens 1 to 3 had excellent slidability and voltage resistance, and also were capable of imparting excellent pressing properties and self-healing properties to a film capacitor. More specifically, the film specimens 1 to 3 had exceptionally good slidability due to the low coefficient of static friction 1 and the low coefficient of static friction 2, and exceptionally good voltage resistance with a dielectric breakdown voltage of 300 V/μm or more, and further were capable of imparting excellent pressing properties and self-healing properties to a film capacitor. Further, the film specimens 1 to 3 had a glass transition point of 130° C. or more, with excellent heat resistance.

(206) The film specimen 4 had an area ratio 2 of less than 0.0998%. Also, in the metallized film state formed by vapor depositing a metal layer on the first main surface of the film specimen 4, the area ratio 2' was less than 0.118%. As a result, the film specimen 4 had a coefficient of static friction 1 and a coefficient of static friction 2 higher than those of the film specimens 1 to 3, and was not capable of imparting excellent pressing properties and self-healing properties to a film capacitor.

(207) The film specimen 5 had an area ratio 2 of more than 1.13%. Also, in the metallized film state formed by vapor depositing a metal layer on the first main surface of the film specimen 5, the area ratio 2' was more than 1.24%. As a result, the film specimen 5 had a lower dielectric breakdown voltage in comparison with the film specimens 1 to 3.

(208) The film specimen 6 had an area ratio 1 of more than 6.04%. Also, in the metallized film state formed by vapor depositing a metal layer on the first main surface of the film specimen 6, the area ratio 1' was more than 6.17%. As a result, the film specimen 6 had a lower dielectric breakdown voltage in comparison with the film specimens 1 to 3.

(209) The film specimen 7 had an area ratio 3 of more than 0.100%. Also, in the metallized film state formed by vapor depositing a metal layer on the first main surface of the film specimen 7, the area ratio 3' was more than 0.100%. As a result, the film specimen 7 had a lower dielectric breakdown voltage in comparison with the film specimens 1 to 3.

(210) The film specimen 8 included a protrusion having an aliphatic isocyanate, and therefore was not capable of imparting excellent self-healing properties to a film capacitor.

(211) The film specimen 9 included no protrusion. Also, in the metallized film state formed by vapor depositing a metal layer on the first main surface of the film specimen 9, there exists no protruding part. As a result, the film specimen 9 had a coefficient of static friction 1 and a coefficient of static friction 2 higher than those of the film specimens 1 to 3, and was not capable of imparting excellent pressing properties and self-healing properties to a film capacitor. In the film specimen 9, the first main surface was gradually concave on the whole, so that the area ratio 1 and the area ratio 2 were not 0%. Also, in the metallized film state formed by vapor depositing a metal

layer on the first main surface of the film specimen 9, the surface of the metal layer was gradually concave on the whole, so that the area ratio 1' and the area ratio 2' were not 0%.

(212) Due to a filler blended therein, the film specimens 10 to 12 had a lower dielectric breakdown voltage in comparison with the film specimens 1 to 3. Also, the film specimen 10 had a coefficient of static friction 1 and a coefficient of static friction 2 higher than those of the film specimens 1 to 3, and was not capable of imparting excellent pressing properties and self-healing properties to a film capacitor.

REFERENCE SIGNS LIST

(213) **10**: Film capacitor **11**: First metallized film **12**: Second metallized film **13**: First dielectric film **13a**: First main surface of first dielectric film **13b**: Second main surface of first dielectric film **14**: Second dielectric film **14a**: First main surface of second dielectric film **14b**: Second main surface of second dielectric film **15**: First metal layer **16**: Second metal layer **40**: Wound body **41**: First external electrode **42**: Second external electrode **61**: Split electrode part **62**: Electrode part **63**: Fuse part **64**: Insulating slit **110**: Film (dielectric film) **110a**: First main surface of film **110b**: Second main surface of film **120**: Protrusion of film **130**: Flat part of film **210**: Metallized film **220**: Metal layer **220a**: Surface of metal layer **230**: Protruding part of metal layer **240**: Flat part of metal layer L: Region of second metallized film M: Length of region of second metallized film N: End side of second metallized film P: Center point of end side of second metallized film Q: Centerline S: Thickness of film T: Lamination direction W: Width direction

Claims

1. A film capacitor comprising: a wound body in which a dielectric film and a metal layer are wound, the dielectric film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, the dielectric film including a first main surface and a second main surface opposed to each other in a thickness direction, the metal layer being disposed at least on the first main surface of the dielectric film, wherein the first main surface of the dielectric film includes a plurality of protrusions having the second organic material, and when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface of the dielectric film, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.
2. The film capacitor according to claim 1, wherein a surface of the metal layer on an opposite side of the first main surface of the dielectric film includes a plurality of protruding parts coinciding with the plurality of the protrusions.
3. The film capacitor according to claim 2, wherein, when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less.
4. The film capacitor according to claim 2, wherein a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film is 1.4 or less.
5. The film capacitor according to claim 2, wherein a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film is 0.2 to 1.4.
6. The film capacitor according to claim 1, wherein the wound body has a flat cross section perpendicular to a scroll axis direction of the wound body.

7. A film capacitor comprising: a wound body in which a dielectric film and a metal layer are wound, the dielectric film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, the dielectric film including a first main surface and a second main surface opposed to each other in a thickness direction, the metal layer being disposed at least on the first main surface of the dielectric film, wherein the first main surface of the dielectric film includes a plurality of protrusions having the second organic material, a surface of the metal layer on an opposite side of the first main surface of the dielectric film includes a plurality of protruding parts coinciding with the plurality of the protrusions, and when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less.
8. The film capacitor according to claim 7, wherein a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film is 1.4 or less.
9. The film capacitor according to claim 7, wherein a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film is 0.2 to 1.4.
10. The film capacitor according to claim 7, wherein the wound body has a flat cross section perpendicular to a scroll axis direction of the wound body.
11. A film comprising a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, and including a first main surface and a second main surface opposed to each other in a thickness direction, wherein the first main surface includes a plurality of protrusions having the second organic material, and when a film face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the first main surface, an area ratio of a region having a height higher than the film face by 0.05 μm or more and less than 0.20 μm is 6.04% or less, an area ratio of a region having a height higher than the film face by 0.20 μm or more and less than 2.50 μm is 0.0998% to 1.13%, and an area ratio of a region having a height higher than the film face by 2.50 μm or more is 0.100% or less.
12. The film according to claim 11, wherein a coefficient of static friction on the first main surface side is 1.0 or less.
13. The film according to claim 11, wherein a coefficient of static friction on the first main surface side is 0.1 to 1.0.
14. A metallized film comprising: a film including a cured product of a first organic material having a hydroxy group and a second organic material that is an aromatic compound having an isocyanate group, and including a first main surface and a second main surface opposed to each other in a thickness direction; and a metal layer disposed at least on the first main surface of the film, wherein the first main surface of the film includes a plurality of protrusions having the second organic material, a surface of the metal layer on an opposite side of the first main surface of the film includes a plurality of protruding parts coinciding with the plurality of the protrusions, and when a metal layer face positioned at an average height in the thickness direction is defined in an area range of 100 μm by 140 μm of the surface of the metal layer, an area ratio of a region having a height higher than the metal layer face by 0.05 μm or more and less than 0.20 μm is 6.17% or less, an area ratio of a region having a height higher than the metal layer face by 0.20 μm or more and less than 2.50 μm is 0.118% to 1.24%, and an area ratio of a region having a height higher than the metal layer face by 2.50 μm or more is 0.100% or less.
15. The metallized film according to claim 14, wherein a coefficient of static friction on a side of the metal layer on the first main surface of the film is 1.4 or less.

16. The metallized film according to claim 14, wherein a coefficient of static friction on a side of the metal layer on the first main surface of the dielectric film is 0.2 to 1.4.
